

Model Predictive Control

Final exam 2021 solution proposal

Problem 1

Tasks

- (i) [20%] Explain and show in detail, how you would formulate nonlinear MPC so that you take into account all the 5 possible values of the parameter GOR. Show your objective function, constraints etc.

Proposed solution:

This problem can be solved with an MPC that utilizes features from multi-objective optimization (MOO) problems. With a nominal MPC, the prediction model uses the nominal values of the uncertain parameter. However, for this process of oil production optimization, five different possible values of the uncertain parameter GOR are defined. The idea here is to construct not just one objective but multiple objective functions by utilizing each possible value of the parameter GOR. A relatively simple method called the weighted sum method can be utilized here.

Let $i = 1, 2, 3, 4, 5$ denote different sets of prediction model for oil flow rate corresponding to each i^{th} value of the parameter GOR as,

$$x_{k+1}^i = f^i(x_k^i, u_k, P_r, P_{wh}^i, P_s, GOR^i) \quad (1)$$

For any i^{th} prediction model, we can define the i^{th} nonlinear objective function for maximizing the oil production (however posed as a minimization problem and hence the negative sign) as,

$$J_i = \sum_{k=1}^N -(x_k^i)^T P (x_k^i) + u_{k-1}^T R u_{k-1} \quad (2)$$

s.t

$$x_{k+1}^i = f^i(x_k^i, u_k, P_r, P_{wh}^i, P_s, GOR^i) \quad (3)$$

$$x_{min} \leq x_k \leq x_{max} \quad (4)$$

$$u_{min} \leq u_k \leq u_{max} \quad (5)$$

The weighting matrices are P and R , and the prediction horizon is denoted by N . Note that, GOR^i denotes the i^{th} value of the parameter GOR at any time k . Now, in order to utilize the weighted sum method for solving MPC problem, the following MOO problem can be formed:

$$\min_{\substack{u_k \\ 0 \leq k \leq N-1}} J = \sum_{i=1}^5 w_i J_i \quad (6)$$

s.t

$$x_{k+1}^i = f^i(x_k^i, u_k, P_r, P_{wh}^i, P_s, GOR^i) \quad (7)$$

$$x_{min} \leq x_k \leq x_{max} \quad (8)$$

$$u_{min} \leq u_k \leq u_{max} \quad (9)$$

Here, w_i is the weight for each objective function. Since there is equal likelihood or probability that the uncertain parameter will take any one of the five different values, the weights w_i can be chosen to be the same for all the objectives. So there is no point/need to favour one objective upon the other.

The main advantage of using MOO for solving this MPC problem is that we only need to optimize N number of unknown variables i.e. we only need to optimize u_k for $0 \leq k \leq N - 1$ even if we consider all 5 sets of GOR values. In other words, we do not need to formulate the MPC problem in such a way that we require $5N$ number of variables to optimize, i.e. N number of variables to optimize for each i^{th} value of the parameter GOR.

- (ii) [5%] Assuming that the constraint on u_{min} is a soft constraint. How would you re-formulate the nonlinear optimal control problem to handle infeasibility that might arise due to constraint on u_{min} . Please also note that the minimum speed of the pump cannot be less than zero (since zero pump speed means that the pump is already completely stopped).

Suggested Solution:

If we consider that the constraint on u_{min} is a soft constraint, then we can use slack variables to relax it in order to avoid infeasibilities. The slack variables can then be added to the list of variables to be optimized i.e. with constraint relaxation, both the slack variables and the control input (pump speed u) should be considered as the decision variables. However, since the minimum speed of the pump cannot be less than zero, we should make sure that during the relaxation of the constraint on u_{min} , it does not become negative. To ensure this, a new inequality constraint ($u_{min} - S_k \geq 0$) can be added. Then the original optimization problem can be re-formulated as,

$\min_{\substack{u_k, S_k \\ 0 \leq k \leq N-1}} J = \sum_{i=1}^5 w_i J_i$	(10)
s.t	
$x_{k+1}^i = f^i(x_k^i, u_k, P_r, P_{wh}^i, P_s, \textcolor{red}{GOR}^i)$	(11)
$x_{min} \leq x_k \leq x_{max}$	(12)
$\textcolor{red}{u}_{min} - S_k \leq u_k \leq u_{max}$	(13)
$\textcolor{red}{u}_{min} - S_k \geq 0$	(14)

Problem 2

Tasks

- (i) [15%] Explain and show in detail how QR factorization can be used to eliminate the equality constraints in order to reduce the size of the optimal control problem. Include all necessary calculations and equations. In addition, please show the detailed intermediate steps of how expressions for $\tilde{H}$ and $\tilde{c}$ are calculated.

Note: For Problem 2, task (i), you cannot just say “after simplifying and re-arranging equation...” we get the expression for H and $\tilde{c}$ and then only write down/copy the final expressions for H and $\tilde{c}$. You should also do and show your own calculations. Students who have only written the final expression for H and $\tilde{c}$ without showing the intermediate calculations/steps will not receive full score.

Suggested solution:

The idea here is to use the linear equality constraint $A_e z = b_e$ present in the original problem to split the unknown variables z into two parts: "basic variables" and "non-basic variables". Using the equality constraint, the basic variables are expressed in terms of the non-basic variables. The basic variables (which are the functions of non-basic variables) are then substituted in the objective function of equation. The objective function will then only have the non-basic variables. This will also result in the elimination of the linear equality constraint & we will obtain the reduced problem.

Let us consider the linear equality constraint as,

$$A_e z = b_e \quad (15)$$

Let us assume equation (15) has n number of equations i.e. it is a compact form of n linear equality constraints. Then, $A_e \in \mathbb{R}^{n \times n_z}$, where n_z is the total number of unknown variables that are listed in vector z . Let the rank of A_e be r i.e. $r = \text{rank}(A_e)$. Rank of a matrix gives you the number of linearly independent rows of the matrix, here in this case, the number of independent linear equality constraints.

Now, let us decompose A_e into the product of Q and R as,

$$A_e = QR$$

where Q = orthogonal matrix i.e. $Q^T Q = I$ and $Q \in \mathbb{R}^{n \times n}$, R = upper triangular matrix, $R \in \mathbb{R}^{n \times n_z}$.

Furthermore, R can be written as,

$$R = \begin{bmatrix} R_1 & R_2 \\ 0 & 0 \end{bmatrix} \quad (16)$$

where, $R_1 \in \mathbb{R}^{r \times r}$ is full rank i.e. $\text{rank}(R_1) = r = \text{rank}(A_e)$ and $R_2 \in \mathbb{R}^{r \times (n_z - r)}$ is the upper right submatrix of R .

Note that the 0's under R_1 has the size of $(n - r) \times r$ and the 0's under R_2 has the size of $(n - r) \times (n_z - r)$. So, we can write the linear equality constraints of equation (15) as,

$$A_e z = b_e$$

$$QRz = b_e$$

Multiplying both sides by Q^T we get,

$$\underbrace{Q^T Q}_I R z = \underbrace{Q^T b_e}_{\bar{b}_e}$$

$$R z = \bar{b}_e \quad (17)$$

Now let us split the vector of unknowns z into $z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$ or $z^T = (z_1^T, z_2^T)$

where, z_1 are the basic variables of size $(r \times 1)$ and z_2 are the nonbasic variables of size $(n_z - r) \times 1$.

Also let us split $\bar{b}_e$ into two parts as,

$$\bar{b}_e = \begin{bmatrix} \bar{b}_{e,1} \\ \bar{b}_{e,2} \end{bmatrix} \quad (18)$$

where, $\bar{b}_{e,1} \in \mathbb{R}^{r \times 1}$ and $\bar{b}_{e,2} \in \mathbb{R}^{(n-r) \times 1}$

Then we have from equations (4.24, 4.25 & 4.26) we get,

$$\begin{bmatrix} R_1 & R_2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} \bar{b}_{e,1} \\ \bar{b}_{e,2} \end{bmatrix} \quad (19)$$

From equation 19 we get,

$$R_1 z_1 + R_2 z_2 = \bar{b}_{e,1}$$

Since R_1 is full ranked, it is invertible. Then we can express z_1 (the basic variables) in terms of z_2 (the non-basic variables) as,

$$z_1 = R_1^{-1} (\bar{b}_{e,1} - R_2 z_2) \quad (20)$$

So, in summary we have,

$$\begin{aligned} z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} &= \begin{bmatrix} R_1^{-1} (\bar{b}_{e,1} - R_2 z_2) \\ z_2 \end{bmatrix} = \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} - R_1^{-1} R_2 z_2 \\ 0 + z_2 \end{bmatrix} \\ &\downarrow \\ z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} &= \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \end{aligned} \quad (21)$$

The sizes of zeros and identity matrix in equation (21) are: $0 \rightarrow (n_z - r) \times 1$ and $I \rightarrow (n_z - r) \times (n_z - r)$.

Now let us substitute z from equation (21) in the original problem given by equation (4.13) i.e.

$\Rightarrow \frac{1}{2} z^T H z + c^T z$ can be written as,

$$\begin{aligned} \frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right)^T H \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) \\ + c^T \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) \end{aligned} \quad (22)$$

We can see that equation (22) has only the non-basic variable z_2 .

Comment: Many students have not shown the intermediate calculations for $\tilde{H}$, $\tilde{c}$ and $\tilde{K}$ as shown by the red text below: from equation (22a) to (22e). It was specifically stated in the exam task description that intermediate calculations are necessary. Students who have not shown intermediate calculations from equation (22a) to (22e), but have only directly shown equation (23) have not received full score.

To solve equation (22) we can make use of the matrix transpose properties. If for example A and B are two matrices then, matrix transpose properties say that $(A + B)^T = A^T + B^T$ and $(AB)^T = B^T A^T$.

Then we have from equation (22),

$$\frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T + z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T \right) H \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) + c^T \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + c^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \quad (22a)$$

We can further solve equation 22(a) as,

$$\frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 + z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) + c^T \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + c^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \quad (22b)$$

Then we can further go on solving 22(b) as,

$$\frac{1}{2} \left(z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) + \frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \right) + c^T \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 + z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \right) + c^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \quad (22c)$$

We know that H is a symmetric diagonal matrix, i.e. $H^T = H$. Then if A and B are any two matrices, then from matrix transpose property, we get, $A^T H B = B^T H A$. Thus the second term of the “green coloured” expression $z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}$ can also be written as $\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2$. Then we have from equation (22c),

$$\frac{1}{2} \left(z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) + \frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \right) + c^T \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 + c^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \quad (22d)$$

From the “green coloured” part of equation 22(d) we have z_2 as common. Then we have,

$$\frac{1}{2} \left(\underbrace{z_2^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2}_{\tilde{H}} \right) + \underbrace{\frac{1}{2} \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \right) + c^T \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}}_{\tilde{K}} + \underbrace{\left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} + c^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} \right)}_{\tilde{C}^T} z_2 \quad (22e)$$

In compact form, equation (22e) can be expressed as,

$$\min_{z_2}, \frac{1}{2} z_2^T \tilde{H} z_2 + \tilde{c}^T z_2 + \tilde{K} \quad (23)$$

where,

$$\tilde{H} = \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} R_2 \\ I \end{bmatrix} \quad (24)$$

$$\tilde{c}^T = \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} + c^T \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}$$

Or by using transpose property of matrices $(A + B)^T = A^T + B^T$ and $(AB)^T = B^T A^T$ (25)

$$\tilde{c} = \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix}^T c$$

$$\tilde{K} = \frac{1}{2} \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix}^T H \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + c^T R_1^{-1} \begin{bmatrix} \bar{b}_{e,1} \\ 0 \end{bmatrix} = \text{constant} \quad (26)$$

Since $\tilde{K}$ is constant, it can be safely removed from the optimization problem. Then the objective of the reduced problem is,

$$\min_{z_2} \frac{1}{2} z_2^T \tilde{H} z_2 + \tilde{c}^T z_2$$

At this stage we have eliminated the equality constraint. But the original optimization problem also has linear inequality constraints and bounds.

For the inequality constraints we have,

$$A_i z \leq b_i$$

Substituting z from equation (21) in this above equation we get,

$$\begin{aligned} A_i \left(\begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \right) &\leq b_i \\ A_i \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 &\leq b_i - A_i \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \end{aligned} \quad (27)$$

For the bounds, we have ,

$$z_L \leq z \leq z_H$$

Substituting z from equation (21) in this above equation we get,

$$z_L \leq \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} + \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \leq z_H$$

$$z_L - \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \leq \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \leq z_H - \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \quad (28)$$

Equation (28) can be written separately using two equations (i.e. bounds can be expressed as two inequality constraints) as,

$$-\begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \leq -z_L + \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \quad (29)$$

$$\begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} z_2 \leq z_H - \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \quad (30)$$

Equations (28), (29) & (30) can be written in compact form as,

$$\tilde{A}_i z_2 \leq \tilde{b}_i$$

where,

$$\tilde{A}_i = \begin{bmatrix} A_i \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} \\ -\begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} \\ \begin{bmatrix} -R_1^{-1} R_2 \\ I \end{bmatrix} \end{bmatrix}, \quad \tilde{b}_i = \begin{bmatrix} b_i - A_i \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \\ -z_L + \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \\ z_H - \begin{bmatrix} R_1^{-1} \bar{b}_{e,1} \\ 0 \end{bmatrix} \end{bmatrix} \quad (31)$$

Thus, the original problem has been formulated as the reduced problem as,

$$\min_{z_2} \frac{1}{2} z_2^T \tilde{H} z_2 + \tilde{c}^T z_2 \quad (32)$$

subject to,

$$\tilde{A}_i z_2 \leq \tilde{b}_i \quad (33)$$

where $\tilde{H}$ is given by equation (24), $\tilde{c}$ is given by equation (25), $\tilde{A}_i$ and $\tilde{b}_i$ are given by equation (31).

The reduced problem given by equations (32 & 33) does not have equality constraints but it is equivalent to the original problem. We can solve the reduced problem of equations (32 & 33) using qpOASES solver in Simulink or quadprog solver in MATLAB.

Let us assume that the optimal values returned by the solver be denoted by z_2^* . After finding the optimal values z_2^* of the reduced optimization problem of (32 & 33), we can find the optimal values of the basic variables as,

$$z_1^* = R_1^{-1} (\bar{b}_{e,1} - R_2 z_2^*) \quad (34)$$

Finally, we have the optimal values of the unknown variables as, $z^* = \begin{bmatrix} z_1^* \\ z_2^* \end{bmatrix}$

- (ii) [5%] Will you use optimization problem given by Equation (5) or that given by Equation (6) to develop an LQ optimal controller for embedded systems? Give proper explanation.

Suggested Solution:

When the size of an optimal control problem is reduced the control problem becomes compact and is usually solved faster by an optimization solver which does not support sparsity. For embedded systems, such a compact optimal control problem is advantageous. This is because of the lack of higher computational power on embedded systems. So the compact is the LQ optimal control problem, the faster it can be solved in an embedded system. Thus the reduced LQ optimal control problem given by Equation (6) should be used.

Problem 3

Tasks

- (i) [5%] Describe the concept about warm start when solving an optimization problem recursively/repeatedly. What is its advantage?

Suggested Solution:

MPC algorithm is a recursive optimizer, i.e. the optimal control problem has to be solved repeatedly using the sliding horizon strategy. It is also known that solving an optimization problem is an iterative procedure, and thus most of the solvers need an initial starting point (a guess usually supplied by the end user). If the initial starting point for the optimizer is already close to the true optimum (usually local optimum) value, then it requires less number of iteration to solve the optimization problem.

To aid this, for solving the optimal control problem (i.e. optimization problem) at the current time step, the initial guess to the optimizer is the solution of the optimization problem obtain at the previous time step. This is also known as warm start.

The advantage with the warm start is that the current optimization problem is solved at the expense of fewer iterations, and hence in overall, MPC is solved much faster.

- (ii) [5%] For a real industrial process, a state feedback MPC will seldom work. Do you agree with this statement? If yes why? If no why?

Suggested solution:

Yes, the statement is true. State feedback MPC is just an ideal case which assumes that all the states of the system are perfectly known or measurable. In a real industrial process, this may not always be the case. Some states might simply be unmeasurable, or even if they are measurable, the sensor readings may be noisy. In other cases, even if the states are measurable, it might require a very expensive sensing mechanism. While designing an MPC, a large resource is already put in developing the dynamic model of the process for prediction. Such a model can also be used for estimation of the unmeasured states. Another advantage of using a state estimator (like the Kalman filter) is to filter out the measured states if they are noisy. Thus whether or not the states are measurable, in a real industrial process, output feedback MPC is inevitable, and only the state feedback MPC is not sufficient.

- (iii) [5%] In practice, during the implementation of MPC for an open loop stable system, the terminal constraints and the terminal cost are usually discarded. Do you agree with this statement? Please justify your answer.

Suggested solution:

Yes, the statement is usually true. We should understand that the reason for having the terminal cost and the terminal constraints is to improve the closed loop stability of MPC. But we also know that the length of the prediction horizon has a direct impact on closed loop stability. In fact, for an infinite long horizon control problem (like the LQR), the closed loop stability is guaranteed. In practice, if the prediction horizon is chosen sufficiently long, stability usually remains intact, at least for open loop stable systems. Thus, in practice, for an open loop stable system, the terminal cost and the terminal constraints are usually discarded to make the implementation of MPC less complex.